

Failing the Fish: A Critical Review of Fish Passage and Ecosystem Disruption at the Xayaburi Dam

The WPC Dragons' Dam Report

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Introduction

Energy is a fundamental resource that has driven modern society and will continue to do so. It powers our homes, transportation, industries and technology. With a growing population across the world, the demand for energy has become increasingly greater. Renewable energy allows for a sustainable form of power to cut risks our infrastructure causes to the environment and us. The transition from fossil fuels to the more reliable renewable energy sources has been ongoing for decades and continues to grow rapidly. New innovations to make energy sources more efficient, reliable and accessible have been at the forefront of technological advancements. There are various renewable energy sources being refined everyday such as wind, solar, and geothermal but one that has stood out for a long time has been hydroelectric energy.

With its easy application all across the world and high efficiency, hydroelectric energy has had a major impact on global use and production accommodating people around the world. The International energy agency (IEA) [1] has shown that out of all renewable energy sources, hydroelectric has been the dominant source across the globe. In 2020, 17% of all global electricity was produced by hydropower and 70% all renewable energy output was done by hydroelectric power. Countries such as China, Brazil, Norway and Canada have all been countries to heavily rely on hydroelectric power. With its long history and easy access for all countries, hydroelectric power seems to be an obvious choice for countries to implement and expand on for future generations. However there has been a noticeable growth to other energy sources and plateau to hydropower which can be attributed to the issues it can pose.

Although hydroelectricity has been dominant in the renewable energy space, the issues it causes have been apparent. The issues major dams pose are not only environmental but social and economical and these have long lasting effects on people and the environment. The development of dams changes the river system and landscape causing damage to the various ecosystems that are dependent on rivers. Furthermore, dams pose issues to the livelihood of communities impacting agriculture, droughts / flooding and transportation. Improving on these issues with dams, we can move towards creating sustainable energy and making the implementation better for the environment and communities.

This report will focus on the Mekong River, which is a major river spanning across various mainland Southeast Asian countries. The Mekong River features 13 major dams and many smaller dams that impose their own effects on the surrounding environment. This report will specifically focus on how fish ladders impact the surrounding region. Specifically, this report will look at the Xayaburi Dam in northwest Laos with a proposed new design to improve the issues the current fish ladder faces. Understanding their role is crucial, as fisheries in this region provide food security and livelihoods for millions of people.

The Mekong River and the Importance of Aquatic Life

The Mekong River Basin is a region full of biodiverse aquatic animals that play a key role for people living in the basin and have a dominant position in comparison to other creatures or animals. Their dominance is amplified by the large amounts of fisheries located along the river, due to the Mekong

having some of the most diverse selection of fish in the world [2]. With the large supply of fish and fisheries, there are many benefits that they bring to the countries that surround the river. This ultimately led them to having a crucial role in the livelihood of not only the people who depend on the Mekong river but also the environment as a whole.

The aquatic fish are a large component that drive up the economy present in many countries around the Mekong River, while also supporting many families in the region that rely on fishing as their main source of income [2]. The amount of income generated for the Mekong area is substantial, with the amount being around 20 - 30 billion dollars. It is also noted that issues brought in from hydroelectric cost around 4 billion dollars in economic losses for the region [2].

To further add to the importance of fish to the region, families that live in the Mekong River depend on fish for food security. In the basin, they account for around 47 - 80% of the protein intake for residents [2]. This is substantial in comparison to people living in places such as the United States, where their predominant source of protein is meats which include chicken and beef [3].

With the many socio economic benefits that aquatic animals bring to the basin, it is only natural to assume that there are many positive environmental changes that come from having many biodiverse fish living in the region. Their contribution to many of the core systems that make up many of the aquatic ecosystems cannot be understated.

Fish are extremely important in water ecosystems, as they allow for nutrients to be recycled throughout [4]. Nutrient cycling is defined by Wikipedia as the process of inorganic and organic matter in creating more matter [4], in this instance taking place in bodies of water and aquatic systems. More specifically, fish ingest food that contains certain elements such as nitrogen and phosphorus. The fish can then excrete or egest these important elements back into the aquatic ecosystem, and are recycled by the aquatic animals [5]. This is an essential process, as animals use these elements to grow and develop. Scarcity of these resources may limit the growth of fish in the ecosystem.

To conclude, fish are incredibly important, and are an essential part of our world. Their contributions help millions of people in the Mekong River Basin, and the residents rely on them to live and survive. In the region, fisheries are a major source of income for most families, and have a huge impact on the economy in the many countries around the basin. Being a huge source of household income, fish also contribute to much of the general diet of people in the Mekong River, and most rely on them for food security. In comparison, some other regions mainly rely on chicken or beef to achieve their protein intake, and do not rely nearly as much on fish. The environmental benefits they bring are also a major component as to why fish are crucial in protecting and keeping safe. They allow for nutrients to be cycled throughout the ecosystem, providing other sea creatures with appropriate nutrients to grow.

General Issues Of Mekong River Dams

Starting in the Tibetan Plateau in China spanning nearly 5000 km all the way to the Gulf of Thailand being the 10th largest river in the world, the Mekong river has been a staple to the agricultural, biodiversity and the economy of the countries it spans. The countries it spans include China, Myanmar,

Thailand, Lao PDR, Cambodia and Vietnam [7]. The river has had a rich history of fishing and has made a major impact to the livelihood of the millions who live there. The river plays an important role for many sectors of the economy and has therefore spawned a successful fishing industry in the region which plays a crucial role in the lives of the 65 million inhabitants of the Mekong river.

Since 1957, the Mekong River Commission (MRC) has been endorsed by the UN to provide attention and care to this important river [8]. Developments have been in the works since the 1970s on small scales, and since the 2000s there has been an increase in development where there are now 13 major dams across the river and many minor ones. Countless others have been planned for the upcoming decades to further expand on this industry [9][10]. These developments are intended to promote economic growth by providing electricity across the region. However, as stated before, there have been issues particularly with the major dams in the Mekong region.

One issue that has been propagated by the expansion of dams across the river has been effects on flooding. Flooding is necessary for the growth of agriculture all across the entire river since it replenishes wetland, irrigated farmland, and provides migration paths for fishes. The introduction of dams in the region has caused irregular flooding season patterns. The change in sediments across the river has been detrimental to the biodiversity of the region which brings another issue [11][12]. With the issues posed by dam construction and failed agriculture, there has been a push away from living near the dams. Since the dams take an immense amount of time, resources and space many indigenous and rural populations face conflicting interests living in the area of newly proposed developments. This can cause social, economic and political tension within the region further dividing populations and causing negative stigma around dams.

Overall there are many issues with the development along the Mekong river and directing attention to these issues not only improve the environmental and agricultural issues they pose but also improve the social and political landscape of the region.

The Xayaburi Dam

The Xayaburi Dam is one of the 13 major dams currently on the Mekong river. The dam is in northwest Laos not far from the capital, Vientiane. The dam first began construction in 2012 and was completed in 2019. It has a capacity of 1285 megawatts and an annual production of 7407 gigawatts helping produce power for the surrounding region of Laos. The infrastructure of this dam has been seen as a battery for southeast asia due to its immense ability to produce power. With its ability to produce more power for the region, the long term economic gain of the dam spiked interest for future investors. [13][14][15]. However, the Xayaburi dam does not come without its issues.

Before the dam was even complete, there were disputes between the human rights violations the dam breached during construction. Additionally, the environmental issues the dam proposed on the region was immense and became an immediate concern. The proposed risks of the dam caused it to not be funded by the World Bank and the Asian Development Bank yet it still gained funding and was completed. It was further estimated that the completed construction of the dam would affect 4000 homes and over 200,000 people due to the changes in the economic and environmental landscape of the region.

Mainly the environment was a big concern since the Mekong river features unique species native to the river and the introduction of damaging infrastructure such as dams can further drive species to extinction.

The environmental concerns of the Xayaburi dam have been apparent since the beginning of construction in 2012. A study was done in April of 2022, PIT tagging 77 *H. malcomi* fishes migrating across 354 km. The span of the river being analyzed came in contact with the Xayaburi dam. By the end of 15 months only 5 fishes reached the end. Although the fish did have to encounter across the river's multiple proposed dams, the only actual dam fully constructed was the Xayaburi dam [16]. Although this study isn't a perfect case on the isolated effects of the Xayaburi dam, it does put into perspective the issues the dam and future dams can pose to fishes in the region. In July 2019, as the Xayaburi Dam in Laos neared completion, the Mekong river experienced record-low water levels, particularly affecting regions in Thailand. A test run was performed at the Xayaburi Dam involving closing floodgates, which further reduced downstream water flow. This action led to water levels dropping by nearly 1.8 meters prompting Thai authorities to request a suspension of the dam's testing phase [17][18].

With all these factors contributing to the decline of the environment there has been a spark to social discourse on the Xayaburi dam. With the previously mentioned issues of fish migration and erosion changes, there has been a push of people away from the regions surrounding the rivers. The land being affected is used for farming, trading, hunting, and in some cases transportation and it has been this way for hundreds of years for rural populations. With an abrupt change to the environment, there has been big backlash on the effects of the Xayaburi. An interview by Radio Free Asia with people living across the Mekong region reflects on the issues they face, "These dams often release water that floods our village and gardens," he said. "Our vegetable gardens on the Mekong river bank are often flooded", agreed a farmer living in the province's Ngeun district, adding that water levels on the river constantly fluctuate [19]. This is just one example highlighting the issues faced by the common people across the Mekong river due to the Xayaburi dam. Moreover, there has been a lack of transparency from the constructors side to the public on the logistics of how the mechanism would work to cause the least amount of damage to the environment. A review done by the WWF in October 2011 months before construction began, the review looked at the claims of the Pöyryreport which outlined the construction of the Xayaburi dam. However, The report noted that it violated at least 8 of the 29 policies for fish ladders set by the MRC and yet didn't reverse these issues [20]. The review by the WWF underscores the lack of attention for the upcoming Xayaburi dam which further brings to light the mismanagement by the team behind the Xayaburi dam.

With all of the evidence brought forward of not only management issues but social issues, the Xayaburi dam has been controversial. The issues posed by the construction of the Xayaburi dam have been documented well for many different facets such as environmental and social concerns. Although the dam itself is meant to promote a better livelihood for the people across the river, the impacts of the dam have been apparent since before construction in 2012. Using this information leads us to propose what we deem to be the specific problems with the current fish ladder and how we plan to improve them.

Proposed Problems

With a perspective on the importance of aquatic life in the Mekong river and how impactful the Mekong river is for its inhabitants we can now look at the specific issues of our report. The Xayaburi dam features a fishway that has multiple issues that affect migration up the river. Our design will mainly cater to the cyprinidae family, which is a family of fish that has carps and minnows which are the most abundant fish in the Mekong river [21]. Firstly, there are problems with the 4 entrances to the fishway of the Xayaburi Dam.

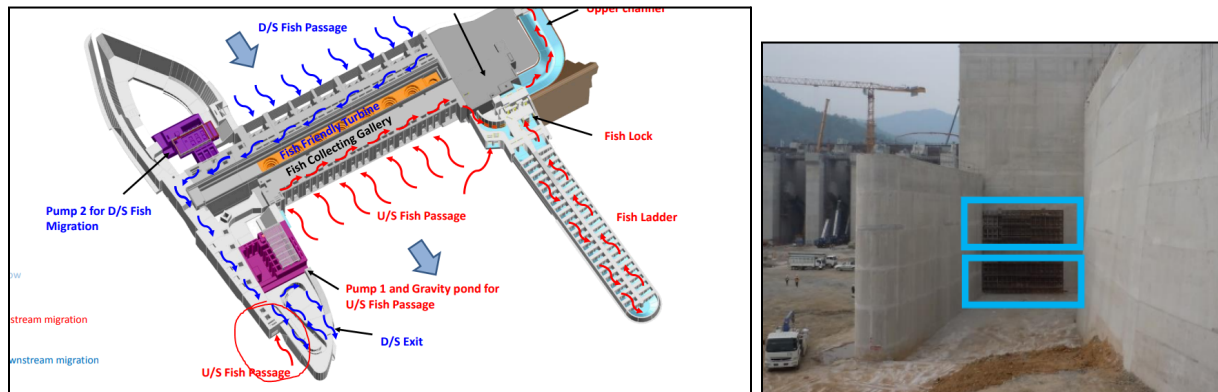


Figure 1.1 & 1.2 - Far left side entrance

Starting at the left upstream passage, there are two gates for fish to enter. For these gates to be useful, they need to be good attraction flows that help guide the fish towards the entrance of the fishpass. Fish tend to “Cue-in” when certain currents are at higher velocities that help them go upstream. A report by Radio Free Asia stated that “out of 223 fishes that were tracked below the dam, only 5 made it to the top” [22]. The way we can combat this problem is to create a larger attraction flow to the entrances. Using siphon styled pumps to create a stronger velocity, will allow the fish to swim towards the entrances but it won't be so fast that it overrules the fish's swimming abilities. However, the velocities are not the only thing needed to attract fish and help guide them. The artificial currents must be set up so that they are roughly parallel ($<30^\circ$) to the current coming down stream [23]. Most entrances should be closer to the shore or where the plant is located, which means that higher velocities need to be used in this case since it is dead center. These attraction flows must also be 70 to 80 percent of the fish's critical speed or burst speed to be able to make sure fish do not fatigue [23]. For this specific entrance at the center of the river, higher speeds must be used to attract fish. There is not much that can be done to this entrance due to the fact that the entrance is unusable during dry season (Nov-April) as there is no water proven in the image below which was taken on January 19, 2024



Figure 1.3 - Dry season at Xayaburi

The second entrance is on the left side of the main power plant (fig 1.4) circled below in fig 1.5. The same criteria needed for the entrance stated above will go for all entrances given.

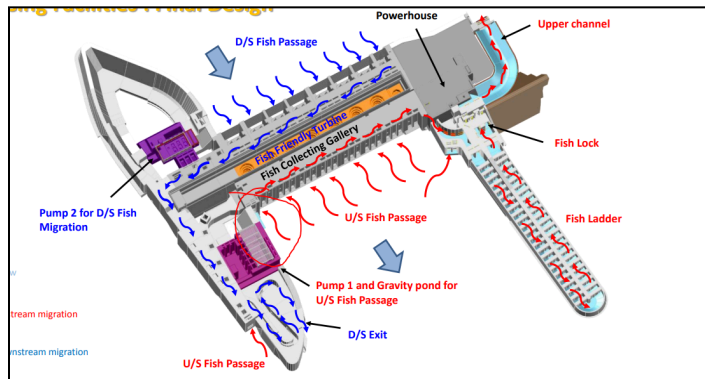


Figure 1.4, 1.5 & 1.6 - Left entrance of main dam

Some design changes based on the water level are needed to improve the functionality of this entrance. Firstly, the water levels are quite low in this part of the dam and are placed at the bottom of a concrete footing which can be impactful against fishes that swim at deeper depths. As seen in the figure 1.6, the water level only gets to about halfway of the highest entrance. Fish swimming near the area may go below the concrete footing and completely miss the exit causing more confusion. The entrance is only at full effect when the tide is high. This also means that the attraction currents used will also be towards the top and not at a full depth. As with the loud noises from the dam itself, fish will tend to stay away from the surface.

The main entrances on the face of the dam are seen below, these entrances are quite well placed and are below the water level at different heights. However, the slots are too small which can cause predatory interactions with multiple fish and more confusion when fish are travelling towards them. If fish run into the dam, they may get disoriented and turn back.



Figure 1.7 & 1.8 - Main entrance

The last entrance of the Xayaburi Dam is on the far right side near the fish ladder as seen below.

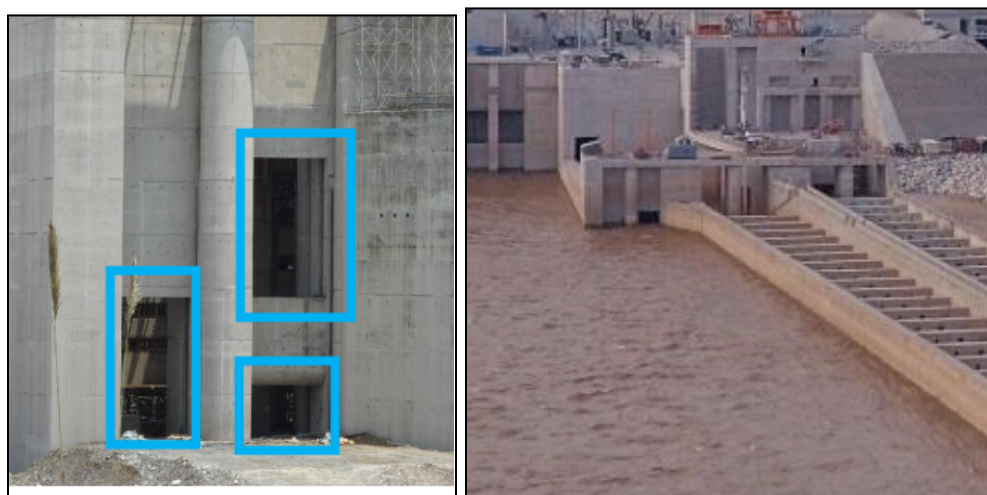


Figure 1.9 & 1.10 - Right side entrance

Similarly to the entrance on the left side of the main front, the water does not cover the entirety of the entrance leading to deep dwelling not being favored for this entrance. Usually bigger fish tend to roam at deeper depths than smaller fish which can create too many predatory interactions leading to certain species being left out of the fishway.



Figure 1.11 & 1.12 - Fish path and Fish lock

Moving onto the fish paths within the dam that transport the fish, there are various issues and proposed problems that cause issues with the fish migration. Firstly, adequate sunlight is needed to make sure the fish are not affected by harsh lighting changes which will limit the flow of the many fish that travel through them, due to the dark concrete channels that have no natural sunlight. One of the channels in figure 1.11 is used to transport fish across the dam, however the lack of sunlight and sharp changes in lighting can stop migrational habits and disorient the fish [24]. This is proven by the department of water affairs in South Africa [24]. This is also the case for when fish pass by turbines or the fish junction shown in fig 1.12. This fish junction is where all the fish are gathered from each entrance. The problem is that the huge amount of barriers and concrete channels limit sunlight and cause fish to be bundled up together creating disorientation and an easy meal for predators.

Moving on to our biggest problem, the fish ladder. There are various issues which affect the migration of fishing moving up the dam. The main problem of the fish ladder is the sunlight and design. The sections of the fish ladder are too tall and the sunlight is very limited. The design is not a well renowned design, and has no proof whether it is good for the cyprinidae family. The top of the slots shown in figure 1.14 are covered with concrete limiting further sunlight. The water level compared to the height of the walls seen in figure 1.13 is quite low, limiting sunlight from all directions. On the floor of the fish ladder there are only small rocks placed which are not natural to the bed of the mekong.

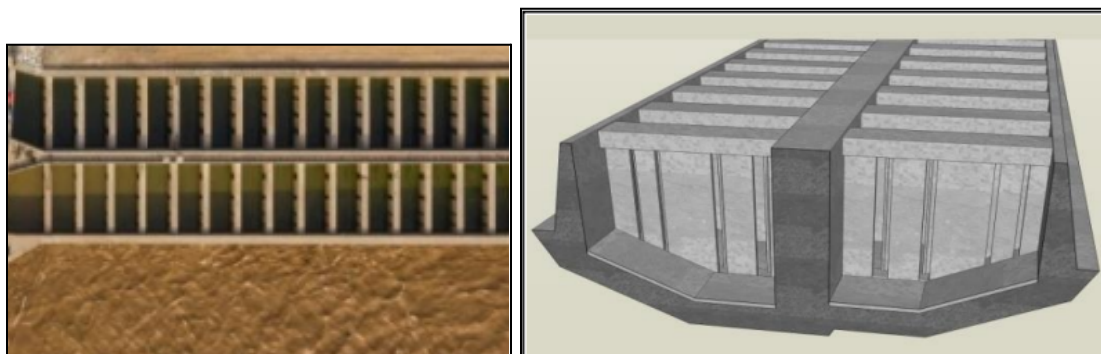


Figure 1.13 & 1.14 - Old design fish gates

One final issue that can be improved upon is with the exit of the fish ladder. Towards the end of the ladder there is a fish elevator which scoops them up. The fish are locked in here for long periods of time which can cause easy predatory behavior.

Solution

We have come up with solutions from all points of the fish way from the entrance to the exit, starting with the entrances. Our new design keeps the base of the original fish ladder but has changes to the design of the slots itself. Stronger pumps to increase the attraction currents velocity, but also making sure the currents are mostly parallel towards downstream current. The tailwater from the turbines can also be used to create a current towards the entrances. Making the exits lower so they are viable during dry season and opening all the entrance slightly larger will allow more fish to exit the dam without any confusion or disorientation. Additionally, always keeping the entrances and pumps open will allow more fish to enter the fishway. Creating adequate sunlight near these sections is also important to remove harsh lighting changes that are known to affect fish migration. Increasing the amount of entrances is also recommended, placing new entrances near the shore or at different elevations will help increase the percentage of fish making it up the river.

The next part that needs some improvements is the fishway to transport fish across the dam. Looking at figure 1.11, there is not adequate sunlight in this section which means within the dam it's safe to assume there is not enough sunlight as well. Many of the fish go through dark concrete pathways and pipes which can be too harsh of a lighting change. Opening up some of these pathways to increase the sunlight or adding artificial sunlight is the solution needed to combat this issue. Looking at figure 1.12 which shows the collection area before the fish ladder. This section is way too confined and lacks real sunlight, opening it up removing the small channels and instead using natural flow to guide the fish, will not only help smaller fish escape predators but also stop harsh lighting changes. Also adding rocks and sand to mimic the habitat of the Mekong region can help make it better for fish travelling within the fishway.

Moving on to the fish ladder itself we have proposed a new design to help aid the fish in completing the ladder. The original fish ladder had issues with sunlight, and other general design issues negatively affecting the migration of the fish. Our design shown in figure 1.15 and 1.16 is a new design inspired by many well renowned designs proven to work in other regions and for the cyprinidae family, the most abundant and important family in the Mekong river [21].

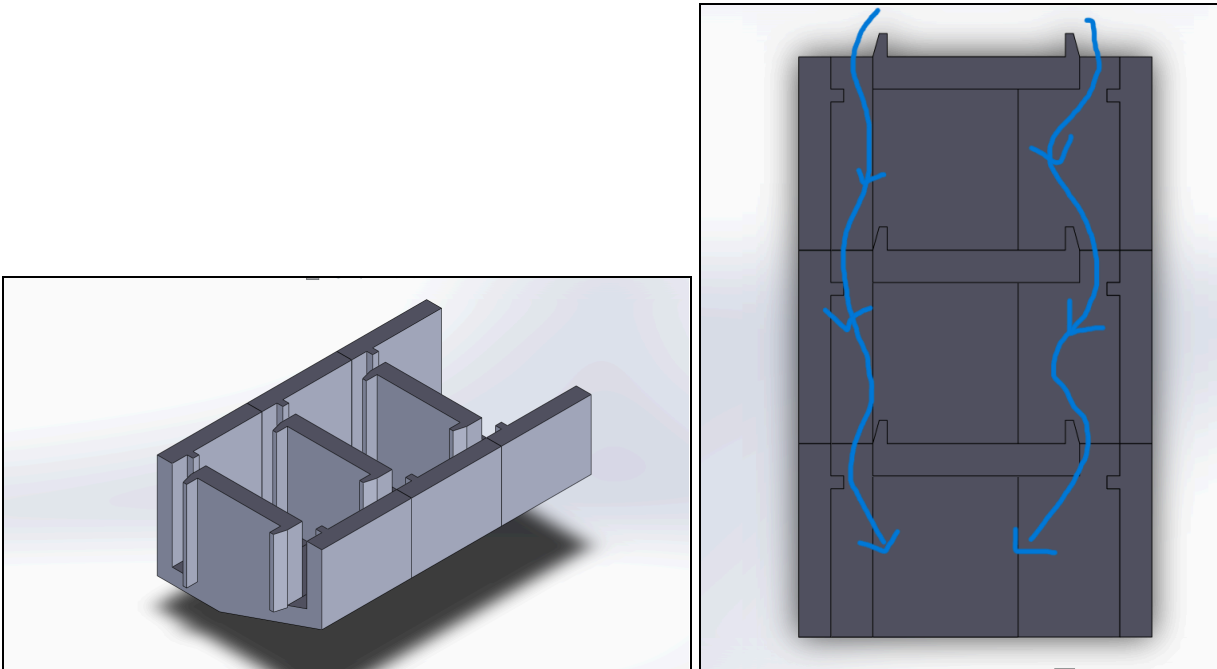


Figure 1.15 & 1.16 - New design

Our design is a modified vertical slotted fish ladder. It is a renowned design recommended for all types of species of fish. Instead of the usual singular vertical slot, we added 2 to increase the amount of fish getting through. Since the slope of the original fish ladder is only 2° of slope, the two slots will not cause high water velocities. Our vertical slot design is great for the cyprinidae family and the large species within the Mekong river. The entrance gaps are 2.5 meters wide and have deflection blocks to help stop current from moving fast and in a straight line. However, due to the low slope some small amount of current will have to be added down the fish ladder to mimic the upstream feeling of the river. The old design had minimal flow and did not align with the migratory habits of fish. Common fish ladders tend to have 10 - 15 degrees of slope [25]. Our design also decreased wall height, removed concrete covering slots to increase sunlight entering the fish ladder. Also adding proper sand, rocks and plants on the bed of the fish ladder to help for waterflow and to simulate the Mekong's river bed which is a proven feature needed for fish ladders [23]. This will help stop fish from getting eaten by predators and create a better fish ladder environment for the fish. This design is not only a very worldwide design known to be great for cyprinids but also great for other species such as the mekong giant catfish which are very important to the Laos people.

Some extra design proposals are to decrease time spent in the fish elevator after the fish ladder to increase the efficiency and to stop bigger fish from eating smaller fish. In addition, adding larger distances between propeller blades to increase the survival rate of the fish is also a great method to increase the migration of fish.

Impact of The Proposed Design

With the solution implemented at the Xayaburi dam, there is an expected restoration of the Mekong river. Our design will expectedly increase the success of long distance migratory fish. One way we can test and see the impact of the success of our design is by using a method that simulates migration to see the relative success [26]. The study was designed for fishes travelling through the Xayaburi dam itself, not the fish ladder. However, using the same methods can help yield similar results to better understand the benefits our design brings. Assuming our design helps increase the success of migratory fish, such as the cyprinidae family, we can now look at the expected impact.

With the understanding of the importance fish bring to the Mekong river, we can expect major benefits for all facets that are effective. Firstly, the environmental benefits will be major since fishes are a staple for the Mekong region. The Mekong is a unique ecosystem home to the second most biodiverse river in the world only behind the Amazon [27]. The balance fishes bring to the life cycle is also crucial, through their feeding and waste, they maintain plant life across the Mekong river which is important for other creatures such as herbivores and intern predators. By reintroducing fishes we can expect all species in the Mekong to benefit, the WWF has highlighted some of the species affected by human activity and we can expect our design to mitigate some of those issues by just introducing fishes back into the ecosystem [28]. Moreover, Mekong fish provide major benefits for humans, as mentioned in section *The Mekong River and the Importance of Aquatic Life*, the people of the region have become dependent on the river and more specifically the fish. Fisheries account for a major sector of the economy in the region, being a multi billion dollar industry. The people also depend on the fishes of the Mekong river as their main source of protein which is one of the three vital nutrients for humans. The overall loss from dams across the Mekong river has been estimated to be 4 billion dollars. Our improvements won't completely reverse the damage that has already been done but as a step in the right direction we can hopefully see a reintroduction to the local communities surrounding the dam and the fishing industry in the region. One area where our design might have flaws is being too steep or having the water velocity be too fast for some larger species. Unique species such as the giant mekong catfish would struggle in fast moving water and need more space for its size. Our design can accommodate for the size but we can't certainly assess the success rate of the giant mekong catfish passing through our design without real life testing, and it should be able to accommodate various species. As mentioned, biodiversity is critical to the Mekong river as it is the second most biodiverse river in the world which is an aspect our design aims to account for.

Our design as a whole isn't meant to be a one solution fix all system, however through our research we have found evidence for general benchmarks we must keep in mind while designing. The ideal impact of our design will promote biodiversity back to the region by being safer and more accessible for various species of fish. With a more biodiverse ecosystem we can expect the local region to refurbish its communities with a bigger fishery industry, bringing even more economic growth and interest to the region. Our design includes minor tweaks, but may possibly provide major benefits for the environment, the people, while also setting an example for other dams across the Mekong river.

Conclusion

In conclusion, the Xayaburi dam, located in Laos atop the Mekong river, has been a major point of conflict. The dam has not only brought concerns about environmental impacts but social clashes with locals feeling impeded by the dam itself. We assessed the biodiverse life within the Mekong river, a

delicate and unique system that houses endangered species and is a vessel for the 65 million inhabitants it supports. The river has long been a source for food, trade, and transportation of the people who live there. With the growing population of mainland southeast Asia, dams have impeded on the natural system of the river and how it intertwined with all life dependent on it. With the introduction of the Xayaburi dam in 2012, the delicate system was then starting to break. Our proposed solution, to not only the Xayaburi dam, but dams across the Mekong river was to improve upon fish ladders to allow for more successful and efficient migration. With the introduction of the Xayaburi dam there has been evidence linking to the decline in migratory fish success which we hoped to fix. Our design aims to provide a safer, easier and more attractive design for fishes to better integrate with the changes we've opposed on them. With our improved design we can predict that there will be an increase in fish population. Dams are here to stay, and contribute much to society today by providing electricity to more remote locations and helping boost economic growth for example, but often affect the environment negatively. Through time we have been successful reaching benchmarks to better accommodate more people, but our efforts can often leave weak points in other aspects of our world. Our duty as human beings for environmental maintenance is a responsibility we carry and should carefully assess in every situation. As we develop and grow further while improving communities across the globe, we still hold the obligation to be stewards of the environment to not only better the lives for other living creatures but also for ourselves.

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